

DEM-stat Anchoring

1. Source and dataset setup

Source: https://ieee-dataport.org/open-access/data-fusion-contest-2019-dfc2019?check_logged_in=1#

Account needed to access data.

Files of Interest: Track 1 / Training data / RGB images 1/1 (Size: 6.59 GB) and Track 1 / Training data / Reference (Size: 8.17 GB)

Location: **data/dfc2019/raw/**

■■■ RGB/Track1-**RGB**
■■■ Truth/Track1-**Truth**

Used RGB(**JAX_004_006_RGB.tif**) and its corresponding AGL(**JAX_004_006_AGL.tif**) from the above folders and checked dimensions, features, height distribution, generated depth-map using DAV2 and calculated Pearson and Spearman correlation.

Result: DAV2 clearly captured useful height-related spatial structure on this WorldView-3 image. Result location: **data/dfc2019/test**

2. Baseline configuration and benchmark subset

- Created **config.json** to store DAV2 configuration. Location: **data/dfc2019/experiments/dav2_baseline/config.json**
- In scripts, created **inspect_dfc2019_pairs.py** to find matching pairs of RGB and AGL. **RGB/AGL** matching pairs: 2783 (JAX: 1014, OMA: 1768)
- In scripts, created **build_dfc2019_manifest.py** to generate a representative 50-tile benchmark subset from the DFC2019 dataset, using AGL data to create diversity in heights. Generated file: **data/dfc2019/experiments/dav2_baseline/manifest.csv**
- In scripts, created **evaluate_dfc2019_dav2.py** that reads **manifest.csv**, executes `run_geotiff()` DAV2 pipeline on each RGB, saves results in **data/dfc2019/experiments/dav2_baseline/depth** as npy files. Then it loads the matching AGLs, flattens both arrays and uses only valid paired pixels, calculates Pearson + Spearman, records runtime and useful diagnostics, write/updates **metrics.csv** located in **data/dfc2019/experiments/dav2_baseline/depth**.

Metric	Mean	Median	Std
Pearson	0.539	0.59	0.219
Spearman	0.446	0.482	0.205

City	Pearson	Spearman
JAX	0.599	0.511
OMA	0.474	0.375

3. Related reports and the jax_022_009 issue

Find related important files in **DepthWizard2/data/dfc2019/experiments/dav2_baseline/reports**: **baseline_summary.json**, **city_summary.csv**, **metrics_source.csv**, **best_worst_tiles.csv**, **tile_ranking.csv**

The **jax_022_009** issue:

It was the lowest-Pearson JAX tile in the 50-tile subset. The RGB and DAV2 maps looked structurally meaningful: buildings, trees, road/canal-like regions and major scene layout were visible. The AGL rendering looked fragmented/noisy.

This raised several hypotheses:

- 1 The AGL visualization might look bad even if the numeric raster were valid.
- 2 Near-ground/zero-height pixels might dominate the correlation.
- 3 Pixel-scale AGL/DAV2 noise or slight registration differences might suppress correlation.
- 4 DAV2 might capture visual structure but fail to preserve true AGL ordering.
- 5 The problem might mainly be scale/calibration rather than relative-depth failure.

For detailed report on the issue and our workflow to diagnose it, read **data/dfc2019/experiments/dav2_baseline/reports/workflow_report.md**

4. Concise workflow

- 1 Created **diagnose_dfc_tile.py**
- 2 Ran on the weak tile **jax_022_009**
- 3 Result in **data/dfc2019/experiments/dav2_baseline/diagnostics/JAX_022_009**: **correlation_diagnostics.csv**, **rgb_dav2_agl_comparison.png**, **dav2_vs_agl_scatter.png**
- 4 Then ran on a strong tile **jax_004_014**: Result in **data/dfc2019/experiments/dav2_baseline/diagnostics/JAX_004_014**: **correlation_diagnostics.csv**, **rgb_dav2_agl_comparison.png**, **dav2_vs_agl_scatter.png**
- 5 Created **diagnose_dfc50.py** and ran it on all 50 tiles. Result in **diagnostic_summary.csv** (Location: **data/dfc2019/experiments/dav2_baseline**)

- 6 Conclusion: Use TWO statistical baselines: global linear calibration and nonlinear monotonic calibration. That is exactly where RGB context, semantic classes, spatial context, and learned fusion become valuable. This justifies our later methods.

5. Calibration setup

Goal: We chose 200 tiles(again, stratified by AGL vertical complexity) for our first calibration. After learning the global linear mapping and global monotonic nonlinear mapping, we'll apply the same to the 50 held-out tiles and calculate MAE/RMSE/Pearson/Spearman.

- In scripts, created **build_calibration_manifest.py** to create the calibration subset. Results in **data/dfc2019/experiments/dav2_calibration/calibration_manifest.csv**
- Note: Because DAv2 is normalized independently per tile, a global cross-tile calibration may fail even if DAv2 is locally useful. That failure would itself be valuable evidence that we need a scene-conditioned/semantic method rather than a global calibration.
- In scripts, created **run_dav2_batch.py**, a DAv2 batch runner to process the 200 calibration tiles cleanly and reuse the same tool later.
- Stored results in depth as spy files and created **manifest.csv** in **data/dfc2019/experiments/dav2_calibration**
- Then we sampled a controlled number of pixels(10,000) from each calibration tile and fit the two mappings. In scripts, created and ran **fit_dav2_calibration.py**
- Results: **linear_mapping.json**, **isotonic_mapping.json**, **isotonic_mapping.joblib** in **data/dfc2019/experiments/dav2_calibration/models**. And **calibration_samples.csv**, **calibration_tile_summary.csv**, **calibration_fit_summary.json** in **data/dfc2019/experiments/dav2_calibration**

6. Findings

For Method 1A: global linear mapping, we got: $\text{height} = 13.2205 \times \text{DAv2} - 1.1372$, $R^2 = 0.1433$ That means the global linear relationship explains only about 14.3% of the variance in AGL on the 200-tile calibration samples. So a single global equation like "nDSM = $13.22 \times \text{DAv2} - 1.14$ " is not a strong metric reconstruction model.

For Method 1B: isotonic, we got 460 thresholds. It means the relationship between DAv2 and AGL is not well represented by one straight line. The isotonic model has learned a fairly detailed monotonic curve. But we cannot judge whether isotonic calibration is actually better yet. We must test both models on the untouched 50-tile benchmark.

DAv2 mean = 0.358, AGL mean = 3.59 m but DAv2 range is always: $0 \rightarrow 1$. It means DAv2 is being normalized independently within each tile. So 0.8 in Tile A does not necessarily represent the same absolute depth/height situation as 0.8 in Tile B. That makes a single global cross-tile mapping fundamentally difficult.